arXiv:2401.15719v5 [math.PR] 29 Aug 2026

# Rates of Convergence in the Central Limit Theorem for Markov Chains, with an Application to TD Learning

R. Srikant
ECE and CSL, UIUC
rsrikant@illinois.edu

September 1, 2026

**Abstract**

We prove a non-asymptotic central limit theorem for vector-valued martingale differences using Stein's method, and use Poisson's equation to extend the result to functions of Markov Chains. We then show that these results can be applied to establish a non-asymptotic central limit theorem for Temporal Difference (TD) learning with averaging.

## 1 Introduction

Starting with the seminal work of (Polyak and Juditsky 1992), where they analyzed the averaging rule of (Polyak 1990, Ruppert 1988), the central limit theorem for martingales has been a valuable tool to understand asymptotic efficiency of stochastic approximation algorithms for optimization. More recently, the central limit theorem for functions of Markov chains has also been used to study the asymptotic efficiency of machine learning algorithms (Borkar et al. 2021, Hu et al. 2024). However, in machine learning applications, finite-time (rather than asymptotic) bounds are preferred to understand the sample complexity of algorithms. However, to the best of our knowledge, there is limited work on the central limit theorems for vector-valued martingales and vector-valued functions of Markov chains that would allow one to both quantify asymptotic efficiency and obtain finite-time bounds for learning and optimization algorithms. A recent paper by (Anastasiou et al. 2019) derives a rate of convergence for the central limit theorem vector-valued martingales; however, the notion of distance between probability distributions in that paper is not sufficient for some applications.

Our goal in this paper is to obtain bounds on the rate of convergence to normality to complement the central limit theorem for Markov chains (Meyn and Tweedie 2012). To this end, we first establish a result for martingale differences by combining the Lindeberg decomposition along with Stein's method, which was first used in (Röllin 2018) for scalar valued martingale differences. This key idea in (Röllin 2018) was extended to the case of vector-valued martingales in (Anastasiou et al. 2019) who also used it to obtain a non-asymptotic version of the central limit theorem in (Polyak and Juditsky 1992). Our approach builds upon the ideas in (Röllin 2018, Anastasiou et al. 2019), but we also introduce new techniques in order to obtain stronger results:

- We would like our results to hold when the deviation from normality is measured using the Wasserstein distance. The results in (Anastasiou et al. 2019) provide bounds only for weaker distances. Therefore, we have to obtain results for more general test functions than the ones considered in (Anastasiou et al. 2019). To do so, we leverage the results on the regularity

of solutions to Stein's equation for vector-valued random variables in (Gallouët et al. 2018, Fang et al. 2019), which makes our analysis considerably different.

- To obtain rates of convergence for the Markov chain central limit theorem, we convert a Markov chain into a martingale difference noise using Poisson's equation; see (Metivier and Priouret 1984, Benveniste et al. 2012, Douc et al. 2018), for example. To the best of our knowledge, this idea has not been used to derive convergence rates for the central limit theorem for functions of Markov chains.

A difference between our results and the results in (Röllin 2018, Anastasiou et al. 2019) for martingales is the fact that we state our convergence bounds in terms of a positive definite matrix $\Sigma_\infty$ which can be viewed as the limiting expected conditional covariance of the martingale differences (see Theorem 1 and the short discussion after the theorem). This is important to us since our goal is to state our results for Markov chains in terms of the asymptotic covariance, which is important in applications; see (Borkar et al. 2021), for example. Therefore, our assumptions are stated differently than in (Röllin 2018, Theorem 1) or in (Anastasiou et al. 2019, Equation 3.1). It is also worth remarking that Stein's method to derive rates of convergence for the martingale central limit theorem is relatively new. The early papers on obtaining rates of convergence for the martingale central limit theorem use other techniques; see, for example, (Bolthausen 1982) and other references in (Röllin 2018). Here, we use Stein's method because we found it easier to quantify the impact of the asymptotic variance on the rate of convergence using Stein's method. Specifically, we show that the regularity results in (Gallouët et al. 2018, Fang et al. 2019) and the calculations in (Röllin 2018, Gallouët et al. 2018) can be used, along with insights from Markov chain theory, to understand the impact of the asymptotic covariance term on the rates of convergence in the martingale and Markov chain central limit theorems.

As an application, we apply the Markov chain central limit theorem to study the distribution of error in TD learning. TD learning is the most widely studied algorithm for evaluating the performance of a given policy from data in Markov Decision Processes (Sutton and Barto 2018). The convergence of TD learning with decaying stepsizes was proved in (Tsitsiklis and Van Roy 1996). Over the last few years, there has been a resurgent interest in understanding the non-asymptotic convergence behavior of TD learning and further using it to study other reinforcement algorithms which use TD learning within their framework. The first result of this type was obtained in (Bhandari et al. 2018) who used a projection step in their version of the TD learning algorithm. The first finite-time bounds for unprojected TD learning were obtained in (Srikant and Ying 2019), who studied both fixed step-sizes and decaying step-sizes. TD learning can be viewed as a special case of linear stochastic approximation with Markovian and multiplicative noise and as a special case of Markovian jump linear systems (Hu and Syed 2019). The results in (Srikant and Ying 2019) were extended to nonlinear, contractive stochastic approximation algorithms in (Chen et al. 2023a), who also derived finite-time performance bounds for specific choices of decaying step-size rules, which also allowed them to study the robustness of different step-size choices. Finite-time performance bounds for variants of TD learning and other schemes have been studied in (Gupta et al. 2019, Doan 2021, 2022) and concentration results for TD learning have been obtained in (Telgarsky 2022, Chen et al. 2023b).

While most of the above papers study standard TD learning, in practice, it is well-known that averaging the iterates can help improve the asymptotic variance. This idea was originally studied for stochastic approximation and stochastic gradient descent in the seminal works in (Ruppert 1988, Polyak 1990, Polyak and Juditsky 1992); see (Moulines and Bach 2011) for a finite-time analysis. More recently, the asymptotic behavior of TD learning and other algorithms with averaging has been studied in (Borkar et al. 2021, Mou et al. 2021, Durmus et al. 2022); see also (Li et al. 2023a) for

related work. As in the papers on stochastic approximation, the conclusion in (Borkar et al. 2021) is that, for a suitable choice of decaying stepsizes, averaging the output leads to efficient estimators in the sense that the variance is minimized. Further, (Borkar et al. 2021) also established functional central limit theorems and central limit theorems for TD learning and other reinforcement learning algorithms. The work in (Mou et al. 2021) shows order-optimal rate of convergence for the mean-squared error, while (Durmus et al. 2022) obtains higher moment bounds and high probability deviation bounds. In contemporaneous work to this paper, finite-time performance bounds on the covariance of the averaged TD learning iterates have been obtained in (Haque et al. 2023) using more robust step-size rules. Their model is more general; they study two-time scale linear stochastic approximation algorithms, with TD learning with averaging being a special case of the models that they study. A central limit theorem for two time-scale stochastic approximation has also been established for two time-scale stochastic approximation in (Hu et al. 2024). Related work also includes a central limit theorem for Polyak-Ruppert averaged Q-learning (Li et al. 2023b).

Our main results in the paper are as follows[1]:

- We derive non-asymptotic central limit theorems for martingales and Markov chains by exploiting ideas from (Röllin 2018, Gallouët et al. 2018, Fang et al. 2019).
- We then extend the results to Markov chains using Poisson's equation to relate Markov chains to martingales. In this case, the asymptotic covariance can be characterized as is well known in the literature; see (Douc et al. 2018), for example.
- Finally, we apply the Markov chain results, along with the ideas in (Polyak and Juditsky 1992), to estimate the rate of convergence in the central limit theorem for TD learning with Polyak-Ruppert averaging. An advantage of the approach here is that our result directly establishes a central limit theorem for the scaled averaged of the iterates, instead of first establishing a function central limit theorem for the entire trajectory of the iterates and then establishing the central limit theorem.
- In addition to the difference in the notion of distance considered here and in (Anastasiou et al. 2019), in the TD learning application, we have to deal with the fact that the noise multiplies the TD learning parameters, whereas the noise is additive in the stochastic gradient application considered in (Anastasiou et al. 2019).

It is important to note that, unlike (Polyak and Juditsky 1992), our result only holds for averaging along with an appropriate decaying step-size rule. Specifically, the result does not hold for fixed step-size TD learning with averaging due to the bias issue pointed out in (Borkar et al. 2021, Lauand and Meyn 2023, Huo et al. 2023)); also see (Nagaraj et al. 2020, Roy and Balasubramanian 2023, Jain et al. 2018) for challenges in dealing with Markov noise in other problems. A nonasymptotic central limit theorem has been established for constant step-size stochastic gradient descent in (Dieuleveut et al. 2020), but the model considered here is different from the problem in (Dieuleveut et al. 2020) (for example, Markovian noise, constant step-size vs decaying step-size) and therefore, the techniques there do not seem to apply to our problem.

The rest of the paper is organized as follows. In Section 2, we establish an upper bound on the rate of convergence in the martingale central limit theorem under some assumptions, and in Section 3, we apply these results to bound the rate of convergence in the central limit theorem for functions of Markov chains using Poisson's equation. The Markov chain results are then applied to

[1]This version of the paper corrects some errors in the version published in Math of OR; while the list of contributions is the same, updated theorems and proofs are presented later in the paper.

study the central limit for averaged iterates of TD learning in Section 4. Concluding remarks and suggestions for future work are provided in Section 5.

# 2 Martingale Central Limit Theorem

In this section, we present a bound on the rate of convergence for the martingale central limit theorem in terms of the Wasserstein distance using Stein's method.

## 2.1 Preliminaries

In this subsection, we will describe the notation we use to establish the rate of convergence in the central limit theorem for vector-valued martingale differences. Let $\{m_k\}_{k\geq 1}$ be a $d$-dimensional martingale difference sequence with respect to a filtration $\{\mathcal{F}_k\}_{k\geq 0}$ i.e., $m_k$ is $\mathcal{F}_k$-adapted, $E(||m_k||) < \infty$, where $||\cdot||$ is the standard Euclidean norm in $\Re^d$ and

$$E(m_k|\mathcal{F}_{k-1}) = 0 \qquad \forall k \geq 1.$$

**Assumption 1.** *Fix $\beta \in (0,1)$. We assume that the martingale difference sequence satisfies the following properties:*

1. *$E\|m_k\|^{2+\beta} < \infty$ for every $k \geq 1$.*
2. *$\Sigma_k = E(m_k m_k^T \mid \mathcal{F}_{k-1})$ exists for every $k \geq 1$.*

$\diamond$

Item (1) in the assumption is motivated by the fact that we want to extend the techniques in (Gallouët et al. 2018) to martingales .

Define $S_n = \sum_{k=1}^n m_k$. We are interested in whether the following normalized sum

$$W_n := \frac{S_n}{\sqrt{n}} \tag{1}$$

converges to $N(0, \Sigma_\infty)$, and if so, we are interested in the rate of convergence. We will use Stein's method to establish such a convergence and to estimate the rate of convergence (Stein 1972, Ross 2011, Chatterjee 2014).

The Wasserstein distance between two $d$-dimensional random vectors $X$ and $Y$ is defined as

$$d_{\mathcal{W}}(X,Y) = \sup_{h \in Lip_1} E[h(X) - h(Y)],$$

where $Lip_L$, the class of Lipschitz functions from $\Re^d$ to $\Re$ with Lipschitz constant $L$, is defined as

$$Lip_L = \{h : |h(x) - h(y)| \leq L||x - y||\}.$$

For later reference, we use the notation $||A||_{op}$ to denote the operator norm of a matrix $A$, i.e.,

$$||A||_{op} = \sup_{x:||x||=1} ||Ax||.$$

The following results from (Gallouët et al. 2018, Propositions 2.2 and 2.3) and (Fang et al. 2019, Theorem 3.1) will be useful to us and so we present them together as a lemma for future reference.

**Lemma 1.** *Let $g \in Lip_L$ and $f$ be the solution to Stein's equation:*

$$g(u) - E(g(Z)) = \Delta f(u) - u^T \nabla f(u), \tag{2}$$

*where $\Delta$ and $\nabla$ are the Laplacian and gradient operators, respectively, and $Z \sim N(0, I)$. Then, $f$ has the following regularity properties for any $\beta \in (0, 1)$:*

1. *$||\nabla^2 f(x) - \nabla^2 f(y)||_{op} \leq \tilde{C}_1(d, \beta)||x - y||^\beta L$, where $\nabla^2$ is the Hessian operator.*
2. *$|\Delta f(x) - \Delta f(y)| \leq \tilde{C}_2(d, \beta)||x - y||^\beta L$, where*

$$\tilde{C}_1 = C_1(d) + \frac{2}{1-\beta}, \quad \tilde{C}_2 = C_2(d) + \frac{2d}{1-\beta}, \quad C_1 = 2^{3/2}\frac{(1+2d)\Gamma((1+d)/2)}{d\Gamma(d/2)}, \quad C_2 = 2\sqrt{\frac{2d}{\pi}}.$$

3. $$\left|\langle \nabla^2 f(x), u_1 u_2^T \rangle_{HS}\right| \leq CL||u_1|| \cdot ||u_2|| \, \forall x, u_1, u_2 \in \Re^d,$$

   *where $C$ is a constant (independent of $d$ and $\beta$) and $\langle \cdot, \cdot \rangle_{HS}$ is the Hilbert-Schmidt inner product (also called the Frobenius inner product), i.e., the sum of the element-by-element product of two matrices.*

◇

Next, we state a well-known result regarding Ornstein-Uhlenbeck processes which will be useful later to establish the nonasymptotic version of the martingale central limit theorem (for example, the result can be easily inferred from the discussions on generators and linear stochastic differential equations in (Arnold 1974)).

**Lemma 2.** *Consider the O-U process*

$$dX_t = AX_t dt + B dw_t,$$

*where $w$ is the standard $d$-dimensional Wiener process, and $A$ and $B$ are $d \times d$ matrices. Then, the generator of the process $\mathcal{A}$ is the operator given by*

$$\mathcal{A}f(x) := \lim_{\delta \to 0} E\left(\frac{f(X_{t+\delta}) - f(X_t)|X_t = x}{\delta}\right) = \nabla^T f(x) Ax + \frac{1}{2} Tr(\nabla^2 f(x) BB^T),$$

*where $f : \Re^d \to \Re$ is any twice differentiable function. If the eigenvalues of $A$ have strictly negative real parts, then the stationary distribution of the O-U process is Gaussian with mean zero and covariance $\Sigma$, which satisfies the Lyapunov equation*

$$\Sigma A^T + A\Sigma + BB^T = 0.$$

*Thus, if $\tilde{Z}_k \sim N(0, \Sigma)$, then $E(\mathcal{A}f(\tilde{Z}_k)) = 0$ for any twice differentiable function $f$.* ◇

It is worth noting that Stein's equation (2) can be motivated by the above lemma when $A = -I$ and $BB^T = 2I$, in which case the right-hand side of (2) is the same as $\mathcal{A}f(u)$. Thus, if $u$ in (2) is replaced by a random vector, then the left-hand side of (2) is a measure of the difference between the distribution of that random vector and a standard Gaussian distribution while the right-hand side will hopefully be small if the distribution of the random vector is close to a standard Gaussian distribution (see, for example, (Barbour 1990, Gotze 1991, Chen et al. 2010)).

## 2.2 Rate of Convergence

We now state and prove the main result of this section.

**Theorem 1.** *Let*

$$V_k := \sum_{j=k}^{n} E(\Sigma_j), \qquad k = 1, \dots, n,$$

*and assume that $V_k$ is positive definite for each $k = 1, \dots, n$. Define*

$$D_k := V_k^{-1/2} E(\Sigma_k) V_k^{-1/2}, \qquad k = 1, \dots, n.$$

*Let $Z \sim N(0, I)$, and let $W_n$ be defined as in (1). Then, under Assumption 1 for the same fixed $\beta \in (0, 1)$,*

$$d_{\mathcal{W}}(W_n, n^{-1/2} V_1^{1/2} Z) \le \frac{1}{\sqrt{n}} \sum_{k=1}^{n} \Bigg[ \tilde{C}_1(d, \beta) \|V_k^{1/2}\|_{op} E\left( \|V_k^{-1/2} m_k\|^{2+\beta} \right) + \tilde{C}_1(d, \beta) \operatorname{Tr}(D_k) \|V_k^{1/2}\|_{op} E\left( \|V_k^{-1/2} m_k\|^{\beta} \right) \Bigg] + \Delta_n,$$

*where*

$$\Delta_n := \sup_{h \in Lip_1} \frac{1}{\sqrt{n}} \left| \sum_{k=1}^{n} E\left[ \operatorname{Tr}\left( A_{k,h}(\mathcal{F}_{k-1}) V_k^{-1/2} \left( E(\Sigma_k) - \Sigma_k \right) V_k^{-1/2} \right) \right] \right|.$$

*Here*

$$A_{k,h}(\mathcal{F}_{k-1}) := E\left[ \nabla^2 f_{k,h}(\widetilde{Z}_k) \mid \mathcal{F}_{k-1} \right],$$

*where $f_{k,h}$ is the $\mathcal{F}_{k-1}$-measurable random solution of the Stein equation defined in the proof below with*

$$a = S_{k-1}, \qquad Q_k = V_k^{1/2}.$$

*and*

$$\widetilde{Z}_k := V_k^{-1/2} T_{k+1} \sim N\left( 0, V_k^{-1/2} V_{k+1} V_k^{-1/2} \right),$$

*with the convention $V_{n+1} = 0$. Moreover,*

$$\|A_{k,h}(\mathcal{F}_{k-1})\|_{op} \le C \|V_k^{1/2}\|_{op}, \quad a.s.,$$

*where $C$ is the constant in statement (3) of Lemma 1. Since $0 \preceq D_k \preceq I$, the same bound also implies the coarser version obtained by replacing $\operatorname{Tr}(D_k)$ by $d$. The sharper form above is used in the Markov-chain applications.*

*Proof.* Fix $h \in Lip_1$. By the standard rescaling of Lipschitz test functions,

$$d_{\mathcal{W}}(W_n, n^{-1/2} V_1^{1/2} Z) = \sup_{h \in Lip_1} \frac{1}{\sqrt{n}} E\left[ h(S_n) - h(T_1) \right],$$

where

$$T_k := \sum_{j=k}^{n} Y_j,$$

and $Y_1, \ldots, Y_n$ are independent Gaussian random vectors, independent of the martingale difference sequence, with

$$Y_j \sim N(0, E(\Sigma_j)).$$

Thus

$$T_k \sim N(0, V_k).$$

The rescaling follows because if $g \in Lip_1$, then $h(x) := \sqrt{n}\, g(x/\sqrt{n})$ also belongs to $Lip_1$, and

$$E\left[g(W_n) - g(n^{-1/2}V_1^{1/2}Z)\right] = \frac{1}{\sqrt{n}} E\left[h(S_n) - h(T_1)\right].$$

Using the Lindeberg decomposition,

$$E\left[h(S_n) - h(T_1)\right] = \sum_{k=1}^{n} E\left[h(S_k + T_{k+1}) - h(S_{k-1} + T_k)\right],$$

where $T_{n+1} = S_0 = 0$.

We now bound a single summand. Fix $k$. Conditional on $\mathcal{F}_{k-1}$, set

$$a = S_{k-1}, \qquad Q_k = V_k^{1/2}.$$

Define

$$\widetilde{h}_k(u) := h(a + Q_k u).$$

Since $h \in Lip_1$, we have

$$\widetilde{h}_k \in Lip_{L_k}, \qquad L_k = \|Q_k\|_{op} = \|V_k^{1/2}\|_{op}.$$

Let $f_k = f_{k,h}$ be the solution of the Stein equation

$$\widetilde{h}_k(u) - E[\widetilde{h}_k(Z)] = \Delta f_k(u) - u^\top \nabla f_k(u).$$

Since $a = S_{k-1}$ is $\mathcal{F}_{k-1}$-measurable, the function $f_k$ is also $\mathcal{F}_{k-1}$-measurable as a random function.

By the Stein equation, with

$$\widetilde{Z}_k := V_k^{-1/2} T_{k+1},$$

we have

$$\begin{aligned} &E\left[h(S_k + T_{k+1}) - h(S_{k-1} + T_k) \mid \mathcal{F}_{k-1}\right] \\ &= E\Big[\Delta f_k\left(V_k^{-1/2}m_k + \widetilde{Z}_k\right) - \left(V_k^{-1/2}m_k + \widetilde{Z}_k\right)^\top \nabla f_k\left(V_k^{-1/2}m_k + \widetilde{Z}_k\right) \mid \mathcal{F}_{k-1}\Big]. \end{aligned}$$

Moreover,

$$\widetilde{Z}_k \sim N(0, \Lambda_k), \qquad \Lambda_k := V_k^{-1/2} V_{k+1} V_k^{-1/2}.$$

Since

$$V_k = E(\Sigma_k) + V_{k+1},$$

we have

$$I - \Lambda_k = V_k^{-1/2} E(\Sigma_k) V_k^{-1/2}.$$

We rewrite the preceding conditional expectation as

$$
\begin{aligned}
&E\left[\mathrm{Tr}\left(\Lambda_k \nabla^2 f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right)\right) - \widetilde{Z}_k^\top \nabla f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right) \mid \mathcal{F}_{k-1}\right] \\
&+ E\left[\mathrm{Tr}\left((I - \Lambda_k)\nabla^2 f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right)\right) \mid \mathcal{F}_{k-1}\right] \\
&- E\left[(V_k^{-1/2} m_k)^\top \nabla f_k(\widetilde{Z}_k) \mid \mathcal{F}_{k-1}\right] \\
&- E\left[(V_k^{-1/2} m_k)^\top \left(\nabla f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right) - \nabla f_k(\widetilde{Z}_k)\right) \mid \mathcal{F}_{k-1}\right].
\end{aligned}
$$

The first term is zero by Gaussian integration by parts: conditional on $\mathcal{F}_k$, the quantity $V_k^{-1/2} m_k$ is fixed and $\widetilde{Z}_k \sim N(0, \Lambda_k)$ is independent of $\mathcal{F}_k$. The third term is zero because $f_k$ is $\mathcal{F}_{k-1}$-measurable as a random function, $\widetilde{Z}_k$ is independent of $\mathcal{F}_k$, and $E[m_k \mid \mathcal{F}_{k-1}] = 0$.

Using the fundamental theorem of calculus, we obtain

$$
\begin{aligned}
&E\left[h(S_k + T_{k+1}) - h(S_{k-1} + T_k) \mid \mathcal{F}_{k-1}\right] \\
&= E\left[\mathrm{Tr}\left(V_k^{-1/2} E(\Sigma_k) V_k^{-1/2} \nabla^2 f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right)\right) \mid \mathcal{F}_{k-1}\right] \\
&\quad - E\left[(V_k^{-1/2} m_k)^\top \left(\nabla^2 f_k(\Theta V_k^{-1/2} m_k + \widetilde{Z}_k) - \nabla^2 f_k(\widetilde{Z}_k)\right)(V_k^{-1/2} m_k) \mid \mathcal{F}_{k-1}\right] \\
&\quad - E\left[(V_k^{-1/2} m_k)^\top \nabla^2 f_k(\widetilde{Z}_k)(V_k^{-1/2} m_k) \mid \mathcal{F}_{k-1}\right],
\end{aligned}
$$

where $\Theta \sim \mathrm{Uniform}[0,1]$ is independent of all other random variables.

Define

$$
A_{k,h}(\mathcal{F}_{k-1}) := E\left[\nabla^2 f_k(\widetilde{Z}_k) \mid \mathcal{F}_{k-1}\right].
$$

Using

$$
\Sigma_k = E[m_k m_k^\top \mid \mathcal{F}_{k-1}],
$$

the last term becomes

$$
-\mathrm{Tr}\left(V_k^{-1/2} \Sigma_k V_k^{-1/2} A_{k,h}(\mathcal{F}_{k-1})\right).
$$

By statement (1) of Lemma 1, the second term above is bounded in absolute value by

$$
\tilde{C}_1(d,\beta) \|V_k^{1/2}\|_{op} E\left[\|V_k^{-1/2} m_k\|^{2+\beta} \mid \mathcal{F}_{k-1}\right].
$$

Therefore,

$$
\begin{aligned}
&E\left[h(S_k + T_{k+1}) - h(S_{k-1} + T_k) \mid \mathcal{F}_{k-1}\right] \\
&\le E\left[\mathrm{Tr}\left(V_k^{-1/2} E(\Sigma_k) V_k^{-1/2} \nabla^2 f_k\left(V_k^{-1/2} m_k + \widetilde{Z}_k\right)\right) \mid \mathcal{F}_{k-1}\right] \\
&\quad - \mathrm{Tr}\left(V_k^{-1/2} \Sigma_k V_k^{-1/2} A_{k,h}(\mathcal{F}_{k-1})\right) \\
&\quad + \tilde{C}_1(d,\beta) \|V_k^{1/2}\|_{op} E\left[\|V_k^{-1/2} m_k\|^{2+\beta} \mid \mathcal{F}_{k-1}\right].
\end{aligned}
$$

Next, add and subtract

$$
\mathrm{Tr}\left(V_k^{-1/2} E(\Sigma_k) V_k^{-1/2} A_{k,h}(\mathcal{F}_{k-1})\right).
$$

This gives

$$\begin{aligned}
&E\left[h(S_k+T_{k+1})-h(S_{k-1}+T_k)\mid \mathcal{F}_{k-1}\right]\\
&\le \operatorname{Tr}\left(A_{k,h}(\mathcal{F}_{k-1})V_k^{-1/2}\left(E(\Sigma_k)-\Sigma_k\right)V_k^{-1/2}\right)\\
&\quad + E\left[\operatorname{Tr}\left(D_k\left[\nabla^2 f_k\left(V_k^{-1/2}m_k+\widetilde{Z}_k\right)-\nabla^2 f_k(\widetilde{Z}_k)\right]\right)\mid \mathcal{F}_{k-1}\right]\\
&\quad + \tilde{C}_1(d,\beta)\|V_k^{1/2}\|_{op}E\left[\|V_k^{-1/2}m_k\|^{2+\beta}\mid \mathcal{F}_{k-1}\right],
\end{aligned}$$

where

$$D_k := V_k^{-1/2}E(\Sigma_k)V_k^{-1/2}.$$

It remains to bound the weighted trace term. Since

$$V_k = E(\Sigma_k)+V_{k+1},$$

we have

$$D_k = I - V_k^{-1/2}V_{k+1}V_k^{-1/2}.$$

Hence

$$0 \preceq D_k \preceq I.$$

Therefore, for any $x,u\in\mathbb{R}^d$,

$$\begin{aligned}
&\left|\operatorname{Tr}\left(D_k\left[\nabla^2 f_k(x+u)-\nabla^2 f_k(x)\right]\right)\right|\\
&\le \operatorname{Tr}(D_k)\left\|\nabla^2 f_k(x+u)-\nabla^2 f_k(x)\right\|_{op}\\
&\le \operatorname{Tr}(D_k)\,\tilde{C}_1(d,\beta)\|V_k^{1/2}\|_{op}\|u\|^{\beta}.
\end{aligned}$$

Applying this bound with

$$x=\widetilde{Z}_k,\qquad u=V_k^{-1/2}m_k,$$

and taking conditional expectation gives

$$\begin{aligned}
&\left|E\left[\operatorname{Tr}\left(D_k\left[\nabla^2 f_k\left(V_k^{-1/2}m_k+\widetilde{Z}_k\right)-\nabla^2 f_k(\widetilde{Z}_k)\right]\right)\mid \mathcal{F}_{k-1}\right]\right|\\
&\le \tilde{C}_1(d,\beta)\operatorname{Tr}(D_k)\|V_k^{1/2}\|_{op}E\left[\|V_k^{-1/2}m_k\|^{\beta}\mid \mathcal{F}_{k-1}\right].
\end{aligned}$$

Thus,

$$\begin{aligned}
&E\left[h(S_k+T_{k+1})-h(S_{k-1}+T_k)\mid \mathcal{F}_{k-1}\right]\\
&\le \operatorname{Tr}\left(A_{k,h}(\mathcal{F}_{k-1})V_k^{-1/2}\left(E(\Sigma_k)-\Sigma_k\right)V_k^{-1/2}\right)\\
&\quad + \tilde{C}_1(d,\beta)\operatorname{Tr}(D_k)\|V_k^{1/2}\|_{op}E\left[\|V_k^{-1/2}m_k\|^{\beta}\mid \mathcal{F}_{k-1}\right]\\
&\quad + \tilde{C}_1(d,\beta)\|V_k^{1/2}\|_{op}E\left[\|V_k^{-1/2}m_k\|^{2+\beta}\mid \mathcal{F}_{k-1}\right].
\end{aligned}$$

Taking expectations, summing over $k$, substituting into the Lindeberg decomposition, and using the rescaling identity for the Wasserstein distance, we obtain, for the fixed $h \in Lip_1$,

$$\begin{aligned}
&E\left[h(W_n) - h(n^{-1/2}V_1^{1/2}Z)\right] \\
&\leq \frac{1}{\sqrt{n}}\sum_{k=1}^{n}\Bigg[\tilde{C}_1(d,\beta)\|V_k^{1/2}\|_{op}E\left(\|V_k^{-1/2}m_k\|^{2+\beta}\right) \\
&\qquad + \tilde{C}_1(d,\beta)\,\mathrm{Tr}(D_k)\|V_k^{1/2}\|_{op}E\left(\|V_k^{-1/2}m_k\|^{\beta}\right)\Bigg] \\
&\quad + \frac{1}{\sqrt{n}}\left|\sum_{k=1}^{n}E\left[\mathrm{Tr}\left(A_{k,h}(\mathcal{F}_{k-1})V_k^{-1/2}\left(E(\Sigma_k) - \Sigma_k\right)V_k^{-1/2}\right)\right]\right|.
\end{aligned}$$

Taking the supremum over $h \in Lip_1$ proves the claimed bound.

Finally, statement (3) of Lemma 1 gives

$$\|A_{k,h}(\mathcal{F}_{k-1})\|_{op} \leq C\|V_k^{1/2}\|_{op}.$$

This completes the proof. □

We note that, if

$$\Sigma_k = \Sigma_\infty \qquad \text{a.s. for all } k,$$

then $\Delta_n = 0$, and the martingale bound reduces to the two Taylor remainder terms.

# 3 Markov Chain Central Limit Theorem

Consider a time-homogeneous Markov chain $\{X_k\}_{k\geq 0}$ on a state space $\mathcal{S}$ with initial distribution $\eta$. Let $r(X_k) \in \Re^d$ denote a vector reward and let $\bar{r} = \pi r$ whenever the chain has invariant distribution $\pi$. We are interested in rates of convergence to normality of the centered additive functional

$$\frac{1}{\sqrt{n}}\sum_{k=1}^{n}\big(r(X_{k-1}) - \bar{r}\big).$$

As is common in the literature, we study this scaled sum by relating the reward process to a martingale via Poisson's equation (Douc et al. 2018, Makowski and Shwartz 2002, Glynn and Meyn 1996, Glynn and Infanger 2023).

## 3.1 Finite State Space Markov Chains

We now show that the martingale theorem can be used to obtain a non-asymptotic CLT for finite-state Markov chains. In Theorem 1, one must control the signed covariance-interaction term. This can be done by applying a second Poisson equation, now to the conditional covariance function.

Throughout this subsection, assume that the Markov chain is stationary, finite-state, and irreducible. Periodicity is allowed. Let

$$g(x) := r(x) - \bar{r}, \qquad \bar{r} := \pi r.$$

Let $V$ solve the Poisson equation

$$V - PV = g.$$

Then

$$g(X_{k-1}) = V(X_{k-1}) - V(X_k) + m_k,$$

where

$$m_k := V(X_k) - E[V(X_k) \mid X_{k-1}]$$

is a martingale difference sequence. Define

$$\Sigma_k := E[m_k m_k^\top \mid \mathcal{F}_{k-1}] = v(X_{k-1}),$$

where

$$v(x) := E[m_1 m_1^\top \mid X_0 = x].$$

Since the chain is stationary,

$$E(\Sigma_k) = \pi(v) =: \Sigma_\infty.$$

Assume throughout this subsection that $\Sigma_\infty$ is positive definite.

We first state the technical estimate needed to control the covariance interaction term.

**Lemma 3.** *Let*

$$V_k := \sum_{j=k}^{n} E(\Sigma_j) = (n-k+1)\Sigma_\infty, \qquad k = 1, \dots, n,$$

*and*

$$H_k := V_k^{-1/2} A_{k,h}(\mathcal{F}_{k-1}) V_k^{-1/2},$$

*where*

$$A_{k,h}(\mathcal{F}_{k-1}) := E[\nabla^2 f_{k,h}(\widetilde{Z}_k) \mid \mathcal{F}_{k-1}]$$

*is the matrix appearing in Theorem 1. Then, for every $\beta \in (0,1)$,*

$$\sup_{h \in Lip_1} \sum_{k=2}^{n} E\|H_k - H_{k-1}\|_{op} \le \frac{C}{(1-\beta)^2} n^{(1-\beta)/2},$$

*where $C$ depends on the chain, $r$, $d$, and $\Sigma_\infty$, but not on $n$ or $\beta$.*

*Proof.* Fix $h \in Lip_1$. Let $c_t := \sqrt{1 - e^{-2t}}$, and let

$$P_A h(x) := E[h(x + A^{1/2} Z)], \qquad Z \sim N(0, I),$$

denote Gaussian convolution with covariance $A$. The Mehler representation of the standard normal Stein solution, followed by differentiation under the integral, gives

$$H_k = -\int_0^\infty e^{-2t} E_{\xi \sim N(0, V_{k+1})}[\nabla^2 P_{c_t^2 V_k} h(S_{k-1} + e^{-t}\xi)]\, dt. \tag{3}$$

The differentiation is justified by the standard Gaussian-smoothing bounds

$$\|\nabla^j P_A h\|_{op} \le C \|A^{-1/2}\|_{op}^{j-1}, \qquad j = 2, 3, 4.$$

Here $\|\cdot\|_{op}$ denotes the usual matrix operator norm for matrices and Hessians, and the multilinear operator norm for high-order derivatives.

The second- and third-derivative bounds imply, respectively,

$$\|\nabla^2 P_A h(x) - \nabla^2 P_A h(y)\|_{op} \le 2C \|A^{-1/2}\|_{op},$$
$$\|\nabla^2 P_A h(x) - \nabla^2 P_A h(y)\|_{op} \le C \|A^{-1/2}\|_{op}^2 \|x - y\|.$$

Since $\min\{1, u\} \le u^\beta$ for $u \ge 0$, interpolation gives

$$\|\nabla^2 P_A h(x) - \nabla^2 P_A h(y)\|_{op} \le C\|A^{-1/2}\|_{op}^{1+\beta}\|x-y\|^\beta, \tag{4}$$

with $C$ uniform over $\beta \in (0,1)$.

Because the state space is finite, $m_k$ is uniformly bounded. Moreover,

$$c(n-k+1)I \preceq V_k \preceq C(n-k+1)I.$$

Using (3), compare $H_k$ and $H_{k-1}$ by changing separately the spatial shift, the covariance inside the Gaussian convolution, and the covariance of the outer Gaussian average. Denote the corresponding contributions by $\Delta_{\text{shift},k}$, $\Delta_{\text{int},k}$, and $\Delta_{\text{ext},k}$.

Since $S_{k-1} - S_{k-2} = m_{k-1}$, (4) gives

$$\begin{aligned} E\|\Delta_{\text{shift},k}\|_{op} &\le CE\|m_{k-1}\|^\beta \int_0^\infty e^{-2t}\|(c_t^2 V_k)^{-1/2}\|_{op}^{1+\beta} dt \\ &\le \frac{C}{1-\beta}(n-k+1)^{-(1+\beta)/2}, \end{aligned}$$

where

$$\int_0^\infty e^{-2t} c_t^{-(1+\beta)} dt = \frac{1}{1-\beta}.$$

For the internal covariance change, stationarity gives $V_{k-1} - V_k = \Sigma_\infty$. Along $A_s = c_t^2(V_k + s\Sigma_\infty)$, the Gaussian-semigroup identity and the fourth-derivative bound yield

$$\begin{aligned} E\|\Delta_{\text{int},k}\|_{op} &\le C\int_0^\infty e^{-2t} c_t^2 \|(c_t^2 V_k)^{-1/2}\|_{op}^3 \|\Sigma_\infty\|_{op}\, dt \\ &\le C(n-k+1)^{-3/2}\int_0^\infty e^{-2t} c_t^{-1} dt \\ &\le C(n-k+1)^{-3/2}. \end{aligned}$$

For the external covariance change, couple

$$\xi_{k-1} = \xi_k + \eta_k, \qquad \xi_k \sim N(0, V_{k+1}), \quad \eta_k \sim N(0, \Sigma_\infty),$$

with independent summands. Applying (4) to the additional shift $e^{-t}\eta_k$ gives

$$E\|\Delta_{\text{ext},k}\|_{op} \le \frac{C}{1-\beta}(n-k+1)^{-(1+\beta)/2}.$$

Consequently,

$$E\|H_k - H_{k-1}\|_{op} \le \frac{C}{1-\beta}(n-k+1)^{-(1+\beta)/2} + C(n-k+1)^{-3/2}.$$

Finally,

$$\sum_{q=1}^{n-1} q^{-(1+\beta)/2} \le \frac{C}{1-\beta} n^{(1-\beta)/2},$$

whereas $\sum_{q\ge1} q^{-3/2} < \infty$. Summing over $k$ and taking the supremum over $h \in Lip_1$ proves the result. □

We can now prove the finite-state Markov-chain CLT.

**Theorem 2.** *Let*

$$U_n = \frac{1}{\sqrt{n}} \sum_{k=1}^{n} (r(X_{k-1}) - \bar{r}).$$

*Assume that the Markov chain is stationary, finite-state, and irreducible, and that* $\Sigma_\infty \succ 0$. *Then, for every* $\beta \in (0,1)$,

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) \le \frac{C}{(1-\beta)^2} n^{-\beta/2},$$

*where* $C$ *does not depend on* $n$ *or* $\beta$. *Consequently,*

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) = O\left(\frac{(\log n)^2}{\sqrt{n}}\right).$$

*Proof.* The reward Poisson equation gives

$$U_n = W_n + \frac{V(X_0) - V(X_n)}{\sqrt{n}}, \qquad W_n := \frac{1}{\sqrt{n}} \sum_{k=1}^{n} m_k.$$

Since $V$ is bounded,

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) \le d_{\mathcal{W}}(W_n, \Sigma_\infty^{1/2} Z) + O(n^{-1/2}).$$

In stationarity,

$$E(\Sigma_k) = \Sigma_\infty, \qquad V_k = (n-k+1)\Sigma_\infty,$$

so the Gaussian target in Theorem 1 is exactly $\Sigma_\infty^{1/2} Z$.

The martingale increments are uniformly bounded, and their moments can be bounded uniformly over $\beta \in (0,1)$. Moreover,

$$\tilde{C}_1(d,\beta) \le \frac{C}{1-\beta} \quad \text{and} \quad \sum_{q=1}^{n} q^{-(1+\beta)/2} \le \frac{C}{1-\beta} n^{(1-\beta)/2}.$$

It follows that the first Taylor remainder in Theorem 1 is at most

$$C(1-\beta)^{-2} n^{-\beta/2}.$$

For the second Taylor remainder,

$$D_k = \frac{1}{n-k+1} I, \qquad \mathrm{Tr}(D_k) = \frac{d}{n-k+1},$$

and the same bound follows.

It remains to control $\Delta_n$. Let

$$B(x) := \Sigma_\infty - v(x).$$

Since $\pi(B) = 0$, there is a bounded matrix-valued solution $\Phi$ of

$$\Phi - P\Phi = B.$$

We may normalize this solution so that $\pi(\Phi) = 0$. Define

$$\Gamma_k := \Phi(X_k) - E[\Phi(X_k) \mid \mathcal{F}_{k-1}].$$

Then

$$E[\Gamma_k \mid \mathcal{F}_{k-1}] = 0,$$

and the Markov property and the Poisson equation give

$$\begin{aligned} B(X_{k-1}) &= \Phi(X_{k-1}) - P\Phi(X_{k-1}) \\ &= \Phi(X_{k-1}) - \Phi(X_k) + \Gamma_k. \end{aligned}$$

Because the chain is stationary,

$$E(\Sigma_k) = \Sigma_\infty, \qquad \Sigma_k = v(X_{k-1}).$$

Hence

$$E(\Sigma_k) - \Sigma_k = \Phi(X_{k-1}) - \Phi(X_k) + \Gamma_k. \tag{5}$$

For fixed $h \in Lip_1$, let

$$H_k := V_k^{-1/2} A_{k,h}(\mathcal{F}_{k-1}) V_k^{-1/2}.$$

Since $H_k$ is $\mathcal{F}_{k-1}$-measurable,

$$E\,\mathrm{Tr}(H_k \Gamma_k) = 0.$$

Using (5) and summation by parts,

$$\begin{aligned} &\sum_{k=1}^{n} E\,\mathrm{Tr}\left(H_k(E(\Sigma_k) - \Sigma_k)\right) \\ &= E\,\mathrm{Tr}(H_1\Phi(X_0)) - E\,\mathrm{Tr}(H_n\Phi(X_n)) \\ &\quad + \sum_{k=2}^{n} E\,\mathrm{Tr}\left((H_k - H_{k-1})\Phi(X_{k-1})\right). \end{aligned}$$

The bound on $A_{k,h}$ in Theorem 1 and the identity $V_k = (n-k+1)\Sigma_\infty$ imply

$$\|H_k\|_{op} \le C(n-k+1)^{-1/2}.$$

Since $\Phi$ is bounded, the two boundary terms are $O(1)$, uniformly over $h \in Lip_1$. Lemma 3 gives

$$\sup_{h \in Lip_1} \sum_{k=2}^{n} E\|H_k - H_{k-1}\|_{op} \le \frac{C}{(1-\beta)^2} n^{(1-\beta)/2}.$$

Consequently,

$$\begin{aligned} \Delta_n &\le \frac{C}{\sqrt{n}} \left(1 + \frac{1}{(1-\beta)^2} n^{(1-\beta)/2}\right) \\ &\le \frac{C}{(1-\beta)^2} n^{-\beta/2}. \end{aligned}$$

Combining the reward boundary estimate, the two Taylor-remainder estimates, and the bound on $\Delta_n$ proves the first assertion. Choosing

$$\beta = 1 - \frac{2}{\log n}$$

for all sufficiently large $n$ proves the second. □

The stationarity assumption can be removed under aperiodicity. We record the extension separately because it is used in the TD-learning application.

**Corollary 1.** *Assume that the Markov chain is finite-state, irreducible, and aperiodic, with an arbitrary initial distribution. If $\Sigma_\infty \succ 0$, then, for every $\beta \in (0,1)$,*

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) \le \frac{C}{(1-\beta)^2} n^{-\beta/2},$$

*with $C$ independent of $n$ and $\beta$, and consequently*

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) = O\left(\frac{(\log n)^2}{\sqrt{n}}\right).$$

*Proof.* Retain the notation from the stationary theorem. The reward Poisson equation again gives

$$U_n = W_n + \frac{V(X_0) - V(X_n)}{\sqrt{n}}, \qquad W_n := \frac{1}{\sqrt{n}} \sum_{k=1}^{n} m_k,$$

and boundedness of $V$ implies

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) \le d_{\mathcal{W}}(W_n, \Sigma_\infty^{1/2} Z) + O(n^{-1/2}).$$

Let

$$V_k := \sum_{j=k}^{n} E(\Sigma_j), \qquad D_k := V_k^{-1/2} E(\Sigma_k) V_k^{-1/2}.$$

Geometric ergodicity gives

$$\|E(\Sigma_k) - \Sigma_\infty\|_{op} \le C\rho^{k-1}$$

for some $\rho \in (0,1)$. Therefore,

$$\frac{1}{n} V_1 = \Sigma_\infty + O(n^{-1}).$$

Since $\Sigma_\infty \succ 0$, the matrix square-root map is locally Lipschitz on the positive-definite cone. Coupling the two Gaussian vectors with the same standard normal vector therefore yields

$$d_{\mathcal{W}}\left(n^{-1/2} V_1^{1/2} Z, \Sigma_\infty^{1/2} Z\right) = O(n^{-1}).$$

Geometric convergence to the positive-definite matrix $\Sigma_\infty$ also implies that, for all sufficiently large $n$, there are constants $0 < c < C < \infty$ such that

$$c(n-k+1) I \preceq V_k \preceq C(n-k+1) I, \qquad k = 1, \dots, n. \tag{6}$$

Indeed, $E(\Sigma_j) \succeq \frac{1}{2}\lambda_{\min}(\Sigma_\infty) I$ for all sufficiently large $j$, while the matrices $E(\Sigma_j)$ are uniformly bounded. For indices $k$ preceding this fixed threshold, the positive-definite tail of the sum still contains a fixed positive fraction of the $n-k+1$ terms.

It follows from (6) that

$$\|V_k^{1/2}\|_{op} \le C(n-k+1)^{1/2}, \qquad \|V_k^{-1/2}\|_{op} \le C(n-k+1)^{-1/2},$$

and

$$\mathrm{Tr}(D_k) = \mathrm{Tr}(V_k^{-1} E(\Sigma_k)) \le \frac{C}{n-k+1}.$$

Since the martingale increments are uniformly bounded, the two Taylor remainders in Theorem 1 are therefore bounded by

$$\frac{C}{(1-\beta)^2} n^{-\beta/2}.$$

It remains to control the covariance-interaction term. Let

$$B(x) := \Sigma_\infty - v(x),$$

and let $\Phi$ be a bounded matrix-valued solution of

$$\Phi - P\Phi = B.$$

Define

$$\widetilde{\Phi}_k := \Phi(X_k) - E[\Phi(X_k)], \qquad \Gamma_k := \Phi(X_k) - E[\Phi(X_k) \mid \mathcal{F}_{k-1}].$$

The Poisson equation gives

$$B(X_{k-1}) = \Phi(X_{k-1}) - \Phi(X_k) + \Gamma_k.$$

Taking expectations and subtracting the resulting identity gives

$$B(X_{k-1}) - E[B(X_{k-1})] = \widetilde{\Phi}_{k-1} - \widetilde{\Phi}_k + \Gamma_k.$$

On the other hand, since $\Sigma_k = v(X_{k-1})$,

$$\begin{aligned} E(\Sigma_k) - \Sigma_k &= E[v(X_{k-1})] - v(X_{k-1}) \\ &= B(X_{k-1}) - E[B(X_{k-1})]. \end{aligned}$$

Therefore,

$$E(\Sigma_k) - \Sigma_k = \widetilde{\Phi}_{k-1} - \widetilde{\Phi}_k + \Gamma_k. \tag{7}$$

Thus, unlike in the stationary theorem, centering is needed because $E[\Phi(X_k)]$ may depend on $k$.

For fixed $h \in Lip_1$, define

$$H_k := V_k^{-1/2} A_{k,h}(\mathcal{F}_{k-1}) V_k^{-1/2}.$$

Because $H_k$ is $\mathcal{F}_{k-1}$-measurable,

$$E\,\mathrm{Tr}(H_k \Gamma_k) = 0.$$

Using (7) and summation by parts gives

$$\begin{aligned} &\sum_{k=1}^{n} E\,\mathrm{Tr}\left(H_k(E(\Sigma_k) - \Sigma_k)\right) \\ &= E\,\mathrm{Tr}(H_1 \widetilde{\Phi}_0) - E\,\mathrm{Tr}(H_n \widetilde{\Phi}_n) \\ &\quad + \sum_{k=2}^{n} E\,\mathrm{Tr}\left((H_k - H_{k-1})\widetilde{\Phi}_{k-1}\right). \end{aligned}$$

The matrices $\widetilde{\Phi}_k$ are uniformly bounded. Moreover, (6) and the bound on $A_{k,h}$ imply

$$\|H_k\|_{op} \le C(n-k+1)^{-1/2},$$

so the two boundary terms are $O(1)$.

The proof of Lemma 3 uses stationarity only through the linear covariance-sum bounds and the bounded one-step covariance increments. Here these ingredients are supplied by (6) together with

$$V_{k-1} - V_k = E(\Sigma_{k-1}), \qquad V_k - V_{k+1} = E(\Sigma_k),$$

whose right-hand sides are uniformly bounded positive-semidefinite matrices. The same spatial-shift, internal-covariance, and external-covariance estimates therefore give

$$\sup_{h \in Lip_1} \sum_{k=2}^{n} E\|H_k - H_{k-1}\|_{op} \le \frac{C}{(1-\beta)^2} n^{(1-\beta)/2}.$$

Consequently,

$$\Delta_n \le \frac{C}{(1-\beta)^2} n^{-\beta/2}.$$

Applying Theorem 1 to $W_n$, comparing its Gaussian target with $\Sigma_\infty^{1/2} Z$, and then using the reward boundary term proves

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) \le \frac{C}{(1-\beta)^2} n^{-\beta/2}.$$

Choosing

$$\beta = 1 - \frac{2}{\log n}$$

for all sufficiently large $n$ gives the stated

$$O\left(\frac{(\log n)^2}{\sqrt{n}}\right)$$

rate. Enlarging $C$ if necessary covers the finitely many remaining values of $n$. □

### 3.2 Extension to General State-Space Markov Chains

We now extend the results to general state-space Markov chains. The ideas are similar to those of the finite state-space case, but we need additional assumptions and work to prove the result. To avoid unnecessary repetition, we use much of the same notation as in the previous section, while additional terminology and machinery are borrowed from (Meyn and Tweedie 2012). Not surprisingly, we need stronger assumptions than the conditions needed in (Meyn and Tweedie 2012) to establish the CLT since we are interested in obtaining the rate of convergence.

**Theorem 3.** *Let $X$ be a $\psi$-irreducible, aperiodic Markov chain on a state space $\mathcal{S}$, with transition kernel $P$ and invariant distribution $\pi$. Suppose there exists a Lyapunov function $\mathscr{V} \ge 1$, unbounded off petite sets, such that $E[\mathscr{V}(X_0)] < \infty$ and*

$$P\mathscr{V}(x) = E[\mathscr{V}(X_{k+1}) \mid X_k = x] \le \lambda \mathscr{V}(x) + L \tag{8}$$

*for some $\lambda \in [0,1)$ and $L < \infty$. Let $r : \mathcal{S} \to \Re^d$ satisfy $\pi\|r\| < \infty$, let $\bar{r} := \pi r$, and define*

$$g := r - \bar{r}, \qquad U_n := \frac{1}{\sqrt{n}} \sum_{k=1}^{n} g(X_{k-1}).$$

*Then the following statements hold.*

1. *Fix $\beta \in (0,1)$. If, for some $C_r < \infty$,*

$$\|g(x)\|^{2+\beta} \le C_r \mathscr{V}(x), \qquad x \in \mathcal{S},$$

*then the Poisson equation $V_r - PV_r = g$ has a solution with the growth bound used below. Define*

$$m_k := V_r(X_k) - E[V_r(X_k) \mid X_{k-1}], \qquad \Sigma_\infty := E_\pi[m_1 m_1^\top].$$

*If $\Sigma_\infty \succ 0$, then*

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) \le \frac{C}{(1-\beta)^2} n^{-\beta/2},$$

*where $Z \sim N(0, I)$ and $C$ does not depend on $n$.*

2. *If*

$$\|g(x)\|^4 \le C_r \mathscr{V}(x), \qquad x \in \mathcal{S},$$

*define $V_r$, $m_k$, and $\Sigma_\infty$ as above. If $\Sigma_\infty \succ 0$, then*

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) = \mathcal{O}\left(\frac{(\log n)^2}{\sqrt{n}}\right).$$

*Proof.* The theorem permits an arbitrary initial distribution satisfying $E[\mathscr{V}(X_0)] < \infty$. We therefore follow the nonstationary argument of Corollary 1, replacing boundedness by weighted moment and weighted Poisson-equation estimates. We first prove the first statement. Set

$$p := 2 + \beta, \qquad a := \frac{1}{p}, \qquad b := \frac{2}{p}.$$

For every $s \in (0,1]$, Jensen's inequality and (8) give

$$P\mathscr{V}^s(x) \le (P\mathscr{V}(x))^s \le \lambda^s \mathscr{V}(x)^s + L^s.$$

The geometric-drift and weighted Poisson-equation results in (Meyn and Tweedie 2012, Chapters 16–17), applied componentwise, therefore imply the following facts. First,

$$|P^j f(x) - \pi f| \le C_s \rho_s^j \|f\|_{\mathscr{V}^s} \big(1 + \mathscr{V}(x)^s\big), \qquad \rho_s \in (0,1), \tag{9}$$

for functions with

$$\|f\|_{\mathscr{V}^s} := \sup_x \frac{|f(x)|}{1 + \mathscr{V}(x)^s} < \infty.$$

Second, the centered reward Poisson equation has a solution satisfying

$$\|V_r(x)\| \le C\big(1 + \mathscr{V}(x)^a\big). \tag{10}$$

The same conclusions apply componentwise to vector- and matrix-valued functions. The drift condition also gives

$$\sup_{j \ge 0} E[\mathscr{V}(X_j)] < \infty. \tag{11}$$

The martingale increments have uniformly bounded $p$th moments. Indeed, using (10) and conditional Jensen's inequality,

$$\begin{aligned} E\|m_k\|^p &\le CE\left[\|V_r(X_k)\|^p + \|E[V_r(X_k) \mid X_{k-1}]\|^p\right] \\ &\le CE\left[1 + \mathscr{V}(X_k) + P\mathscr{V}(X_{k-1})\right] \le C. \end{aligned}$$

Moreover, (10) and (11) imply

$$\sup_{j\geq 0} E\|V_r(X_j)\| < \infty.$$

Consequently,

$$U_n = W_n + \frac{V_r(X_0) - V_r(X_n)}{\sqrt{n}}, \qquad W_n := \frac{1}{\sqrt{n}} \sum_{k=1}^{n} m_k,$$

and the reward boundary term contributes only $\mathcal{O}(n^{-1/2})$ in Wasserstein distance.

Define the conditional covariance function

$$v(x) := E[m_1 m_1^\top \mid X_0 = x], \qquad \Sigma_k = v(X_{k-1}).$$

The growth estimate (10) and conditional Jensen's inequality yield

$$\|v(x)\|_{op} \leq C\big(1 + \mathscr{V}(x)^b\big). \tag{12}$$

In particular,

$$\Sigma_\infty = \pi v$$

is finite. By (9), after integrating with respect to the law of $X_0$ and using $E[\mathscr{V}(X_0)] < \infty$,

$$\|E(\Sigma_k) - \Sigma_\infty\|_{op} \leq C\rho^k \tag{13}$$

for some $\rho \in (0, 1)$.

Let

$$V_k := \sum_{j=k}^{n} E(\Sigma_j), \qquad q_k := n - k + 1.$$

Since $\Sigma_\infty \succ 0$, the geometric convergence in (13) and the uniform upper bound following from (12) imply that, for all sufficiently large $n$,

$$cq_k I \preceq V_k \preceq Cq_k I, \qquad k = 1, \ldots, n. \tag{14}$$

For the lower bound, choose $J$ such that

$$E(\Sigma_j) \succeq \tfrac{1}{2}\lambda_{\min}(\Sigma_\infty) I, \qquad j \geq J.$$

If $k \geq J$, the lower bound follows term by term. If $k < J$, the positive tail from $J$ to $n$ still contains a fixed positive fraction of the $q_k$ terms once $n$ is sufficiently large. The upper bound follows from (12) and (11). Enlarging the constants, if necessary, covers the finitely many remaining values of $n$. Furthermore,

$$\frac{1}{n} V_1 = \Sigma_\infty + \mathcal{O}(n^{-1}). \tag{15}$$

Since $\Sigma_\infty \succ 0$, the matrix square-root map is locally Lipschitz on the positive-definite cone. Coupling the two Gaussian vectors with the same standard normal vector therefore gives

$$d_{\mathcal{W}}\left(n^{-1/2} V_1^{1/2} Z, \Sigma_\infty^{1/2} Z\right) = \mathcal{O}(n^{-1}).$$

We now apply Theorem 1 to $W_n$. From (14),

$$\|V_k^{1/2}\|_{op} \leq Cq_k^{1/2}, \qquad \|V_k^{-1/2}\|_{op} \leq Cq_k^{-1/2}.$$

Also,

$$\mathrm{Tr}(D_k) = \mathrm{Tr}\big(V_k^{-1} E(\Sigma_k)\big) \le \frac{C}{q_k},$$

because $E\|\Sigma_k\|_{op}$ is uniformly bounded. The uniform $p$th moment bound on $m_k$ gives

$$E\|V_k^{-1/2} m_k\|^p \le C q_k^{-p/2}, \qquad E\|V_k^{-1/2} m_k\|^\beta \le C q_k^{-\beta/2}.$$

Since $\widetilde{C}_1(d,\beta) \le C/(1-\beta)$, both Taylor remainders in Theorem 1 are bounded by a constant times

$$\frac{1}{\sqrt{n}} \frac{1}{1-\beta} \sum_{k=1}^{n} q_k^{-(1+\beta)/2} \le \frac{C}{(1-\beta)^2} n^{-\beta/2},$$

where

$$\sum_{q=1}^{n} q^{-(1+\beta)/2} \le \frac{C}{1-\beta} n^{(1-\beta)/2}.$$

It remains to control the covariance-interaction term. As in Corollary 1, define

$$B(x) := \Sigma_\infty - v(x).$$

Then $\pi(B) = 0$, and (12) implies that the matrix-valued Poisson equation

$$\Psi - P\Psi = B \tag{16}$$

has a componentwise solution satisfying

$$\|\Psi(x)\|_{op} \le C\big(1 + \mathscr{V}(x)^b\big). \tag{17}$$

Define

$$\widetilde{\Psi}_k := \Psi(X_k) - E[\Psi(X_k)], \qquad \Gamma_k := \Psi(X_k) - E[\Psi(X_k) \mid \mathcal{F}_{k-1}].$$

The time-dependent centering is needed here because the theorem does not assume that $X_0 \sim \pi$. By the Markov property and (16),

$$\begin{aligned} B(X_{k-1}) &= \Psi(X_{k-1}) - P\Psi(X_{k-1}) \\ &= \Psi(X_{k-1}) - \Psi(X_k) + \Gamma_k. \end{aligned}$$

Taking expectations and subtracting the resulting identity gives

$$B(X_{k-1}) - E[B(X_{k-1})] = \widetilde{\Psi}_{k-1} - \widetilde{\Psi}_k + \Gamma_k.$$

Since $\Sigma_k = v(X_{k-1})$,

$$E(\Sigma_k) - \Sigma_k = B(X_{k-1}) - E[B(X_{k-1})].$$

Therefore,

$$E(\Sigma_k) - \Sigma_k = \widetilde{\Psi}_{k-1} - \widetilde{\Psi}_k + \Gamma_k. \tag{18}$$

For fixed $h \in Lip_1$, let

$$H_k := V_k^{-1/2} A_{k,h}(\mathcal{F}_{k-1}) V_k^{-1/2}.$$

Because $H_k$ is $\mathcal{F}_{k-1}$-measurable and $E[\Gamma_k \mid \mathcal{F}_{k-1}] = 0$,

$$E\,\mathrm{Tr}(H_k \Gamma_k) = 0.$$

Using (18) and summation by parts gives

$$\begin{aligned}
&\sum_{k=1}^{n} E\,\mathrm{Tr}\big(H_k(E(\Sigma_k)-\Sigma_k)\big)\\
&\quad = E\,\mathrm{Tr}(H_1\widetilde{\Psi}_0) - E\,\mathrm{Tr}(H_n\widetilde{\Psi}_n)\\
&\qquad + \sum_{k=2}^{n} E\,\mathrm{Tr}\big((H_k-H_{k-1})\widetilde{\Psi}_{k-1}\big). \qquad (19)
\end{aligned}$$

The Stein bound in Theorem 1 and (14) imply

$$\|H_k\|_{op} \le C q_k^{-1/2}.$$

Together with (17) and (11), this shows that the two boundary terms in (19) are $\mathcal{O}(1)$, uniformly over $h \in Lip_1$.

For the remaining sum, repeat the three-part comparison in Lemma 3. Write the contributions to $\|H_k - H_{k-1}\|_{op}$ from the internal smoothing covariance, the outer Gaussian covariance, and the spatial shift as $\Delta_{\mathrm{int},k}$, $\Delta_{\mathrm{ext},k}$, and $\Delta_{\mathrm{shift},k}$, respectively. Since

$$V_{k-1} - V_k = E(\Sigma_{k-1}), \qquad V_k - V_{k+1} = E(\Sigma_k),$$

and these deterministic covariance increments are uniformly bounded, the fourth-derivative smoothing estimate gives

$$E\left[\Delta_{\mathrm{int},k}\big(1+\mathscr{V}(X_{k-1})^b\big)\right] \le C q_k^{-3/2}.$$

Coupling the outer Gaussian variables by an independent Gaussian increment with covariance $E(\Sigma_k)$ and using the fractional smoothing estimate gives

$$E\left[\Delta_{\mathrm{ext},k}\big(1+\mathscr{V}(X_{k-1})^b\big)\right] \le \frac{C}{1-\beta} q_k^{-(1+\beta)/2}.$$

Finally, since $S_{k-1} - S_{k-2} = m_{k-1}$, the spatial-shift term satisfies

$$\begin{aligned}
&E\left[\Delta_{\mathrm{shift},k}\big(1+\mathscr{V}(X_{k-1})^b\big)\right]\\
&\quad \le \frac{C}{1-\beta} q_k^{-(1+\beta)/2} E\left[\|m_{k-1}\|^\beta\big(1+\mathscr{V}(X_{k-1})^b\big)\right].
\end{aligned}$$

The final expectation is uniformly bounded. Indeed, since $p-\beta = 2$ and $bp/(p-\beta) = 1$, Hölder's inequality gives

$$\begin{aligned}
E\left[\|m_{k-1}\|^\beta \mathscr{V}(X_{k-1})^b\right] &\le (E\|m_{k-1}\|^p)^{\beta/p}\left(E\mathscr{V}(X_{k-1})^{bp/(p-\beta)}\right)^{(p-\beta)/p}\\
&\le C.
\end{aligned}$$

Consequently,

$$\begin{aligned}
&\sum_{k=2}^{n} E\left[\|H_k - H_{k-1}\|_{op}\big(1+\mathscr{V}(X_{k-1})^b\big)\right]\\
&\qquad \le C\sum_{q=1}^{n-1} q^{-3/2} + \frac{C}{1-\beta}\sum_{q=1}^{n-1} q^{-(1+\beta)/2}\\
&\qquad \le \frac{C}{(1-\beta)^2} n^{(1-\beta)/2}.
\end{aligned}$$

By (17), the same bound controls the last sum in (19). After multiplying by the global $n^{-1/2}$ factor in Theorem 1, we obtain

$$\Delta_n \leq \frac{C}{(1-\beta)^2} n^{-\beta/2}.$$

Combining this estimate with the two Taylor remainders, the reward boundary term, and the Gaussian-target comparison proves the first statement.

For the second statement, the fourth-moment condition gives a reward Poisson solution with the fixed growth bound

$$\|V_r(x)\| \leq C\big(1 + \mathscr{V}(x)^{1/4}\big),$$

uniformly bounded fourth moments of $m_k$, and a covariance function $v$ and covariance Poisson solution $\Psi$ with the fixed growth bound

$$\|v(x)\|_{op} + \|\Psi(x)\|_{op} \leq C\big(1 + \mathscr{V}(x)^{1/2}\big).$$

Repeating the preceding argument for any $\beta \in (0,1)$ gives

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) \leq \frac{C}{(1-\beta)^2} n^{-\beta/2},$$

with $C$ uniform in $\beta$. In particular, the weighted spatial-shift estimate is uniform because Hölder's inequality gives

$$E\left[\|m_{k-1}\|^\beta \mathscr{V}(X_{k-1})^{1/2}\right] \leq \left(E\|m_{k-1}\|^4\right)^{\beta/4} \left(E\mathscr{V}(X_{k-1})^{2/(4-\beta)}\right)^{(4-\beta)/4},$$

and $2/(4-\beta) \leq 2/3$. Choosing

$$\beta = 1 - \frac{2}{\log n}$$

for all sufficiently large $n$ gives

$$d_{\mathcal{W}}(U_n, \Sigma_\infty^{1/2} Z) = \mathcal{O}\left(\frac{(\log n)^2}{\sqrt{n}}\right).$$

Enlarging the implied constant covers the finitely many remaining values of $n$. □

# 4 An Application to TD Learning

Temporal Difference (TD) Learning is a common method to learn the performance of a policy in reinforcement learning; see (Bertsekas and Tsitsiklis 1996, Sutton and Barto 2018) for details. In (Tsitsiklis and Van Roy 1996), it was shown that TD learning can be represented as a jump Markovian linear dynamical system as follows:

$$\theta_{k+1} = \theta_k - \epsilon_k(A(X_k)\theta_k + b(X_k)), \tag{20}$$

where $\theta_k \in \Re^d$, $\epsilon_k$ is the stepsize, $\{X_k\}_{k\geq 0}$ is a finite state-space Markov chain, $A(X_k)$ is a $d \times d$ matrix, $b(X_k)$ is a $d$-dimensional vector and $\theta_0$ is a fixed vector. We make the following assumptions.

**Assumption 2.** *We assume that the TD learning algorithm satisfies the following conditions:*

1. *$\{X_k\}$ is an irreducible, aperiodic, finite state Markov chain.*

2. *Let $\bar{A} := E_{X_k \sim \pi}(A(X_k))$. Then, $-\bar{A}$ is a Hurwitz matrix, i.e., all eigenvalues have strictly negative real parts.*

$\diamond$

The first item in the assumption is standard in the reinforcement learning literature (Sutton and Barto 2018), while the second item has been established for TD learning in (Tsitsiklis and Van Roy 1996).

Define $\theta^*$ by $\bar{A}\theta^* + \bar{b} = 0$, where $\bar{b} = E_{X_k \sim \pi}(b(X_k))$. Here, we are interested in the Polyak-Ruppert averaged version of the TD-learning algorithm, i.e.,

$$\bar{\theta}_n = \frac{1}{n} \sum_{k=1}^{n} \theta_k.$$

Our main result of this section characterizes the rate of convergence of this algorithm for an appropriately chosen $\{\epsilon_k\}$. As in the previous theorem, we have chosen to write the result using the $\mathcal{O}$ notation rather than making the constants explicit, but these constants can be made explicit by following the details of the proof of the theorem.

Define the centered TD noise

$$r_{\mathrm{TD}}(x) := (A(x) - \bar{A})\theta^* + (b(x) - \bar{b}). \tag{21}$$

Since $\bar{A}\theta^* + \bar{b} = 0$, we have $\pi r_{\mathrm{TD}} = 0$. Let $V_{\mathrm{TD}}$ solve

$$V_{\mathrm{TD}} - PV_{\mathrm{TD}} = r_{\mathrm{TD}},$$

and define

$$m_{k+1} := V_{\mathrm{TD}}(X_{k+1}) - E[V_{\mathrm{TD}}(X_{k+1}) \mid X_k], \qquad \Sigma_\infty := E_\pi[m_1 m_1^\top]. \tag{22}$$

**Theorem 4.** *Suppose Assumption 2 holds, $\Sigma_\infty \succ 0$, and*

$$\epsilon_k = (k+1)^{-\delta}, \qquad \delta \in (1/2, 1).$$

*Then, for $Z \sim N(0, I)$,*

$$d_{\mathcal{W}}\Big(\sqrt{n}(\bar{\theta}_n - \theta^*), (\bar{A}^{-1}\Sigma_\infty \bar{A}^{-T})^{1/2} Z\Big) = \mathcal{O}\left(\max\left\{\frac{\sqrt{\log n}}{n^{\delta - 1/2}}, \frac{1}{n^{(1-\delta)/2}}\right\}\right).$$

*Proof.* Let $\Delta_k := \theta_k - \theta^*$. From (20),

$$\Delta_{k+1} = (I - \epsilon_k \bar{A})\Delta_k - \epsilon_k (A(X_k) - \bar{A})\Delta_k - \epsilon_k r_{\mathrm{TD}}(X_k).$$

Iterating this recursion, summing over $k = 1, \ldots, n$, and interchanging the order of summation gives

$$\begin{aligned}
\sqrt{n}(\bar{\theta}_n - \theta^*) = \frac{1}{\sqrt{n}} \sum_{k=1}^{n} \prod_{l=0}^{k-1} (I - \epsilon_l \bar{A})\Delta_0 \\
- \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} \epsilon_j \sum_{k=j+1}^{n} \prod_{l=j+1}^{k-1} (I - \epsilon_l \bar{A})(A(X_j) - \bar{A})\Delta_j \\
- \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} \epsilon_j \sum_{k=j+1}^{n} \prod_{l=j+1}^{k-1} (I - \epsilon_l \bar{A}) r_{\mathrm{TD}}(X_j),
\end{aligned}$$

where an empty matrix product is the identity. For $0 \le j \le n$, define

$$\begin{aligned}\Upsilon_j^n &:= \epsilon_j \sum_{k=j+1}^{n} \prod_{l=j+1}^{k-1} (I - \epsilon_l \bar{A}) - \bar{A}^{-1}, \\ \Phi_j^n &:= \epsilon_j \sum_{k=j}^{n-1} \prod_{l=j}^{k-1} (I - \epsilon_l \bar{A}) - \bar{A}^{-1}, \qquad 0 \le j \le n-1.\end{aligned} \tag{23}$$

Thus $\Upsilon_n^n = -\bar{A}^{-1}$. Directly from the definitions,

$$\Upsilon_j^n = \frac{\epsilon_j}{\epsilon_{j+1}} \Phi_{j+1}^{n+1} + \left( \frac{\epsilon_j}{\epsilon_{j+1}} - 1 \right) \bar{A}^{-1}, \qquad 0 \le j \le n-1. \tag{24}$$

Appendix B proves

$$\sup_{n \ge 1} \max_{0 \le j \le n} \|\Upsilon_j^n\|_{op} < \infty, \qquad \frac{1}{n} \sum_{j=0}^{n-1} \|\Upsilon_j^n\|_{op}^2 = \mathcal{O}\Big( n^{-(1-\delta)} \Big). \tag{25}$$

Using $\epsilon_j \sum \prod = \bar{A}^{-1} + \Upsilon_j^n$, we obtain the five-term decomposition

$$\sqrt{n}(\bar{\theta}_n - \theta^*) = \frac{1}{\sqrt{n}} \sum_{k=1}^{n} \prod_{l=0}^{k-1} (I - \epsilon_l \bar{A}) \Delta_0 \tag{26}$$

$$- \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} \bar{A}^{-1} (A(X_j) - \bar{A}) \Delta_j \tag{27}$$

$$- \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} \Upsilon_j^n (A(X_j) - \bar{A}) \Delta_j \tag{28}$$

$$- \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} \Upsilon_j^n r_{\mathrm{TD}}(X_j) \tag{29}$$

$$- \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} \bar{A}^{-1} r_{\mathrm{TD}}(X_j). \tag{30}$$

Denote the five terms on the right-hand side of (26)–(30) by $T_{1,n}, \dots, T_{5,n}$, respectively. We bound the first four in mean norm and apply the Markov-chain CLT to the fifth.

**Step 1: term** (26)**.** Because $\epsilon_0 = 1$ and all factors are polynomials in $\bar{A}$,

$$\epsilon_0 \sum_{k=1}^{n} \prod_{l=0}^{k-1} (I - \epsilon_l \bar{A}) = (I - \epsilon_0 \bar{A})(\Upsilon_0^n + \bar{A}^{-1}).$$

The uniform bound in (25) therefore shows that the mean norm of (26) is $\mathcal{O}(n^{-1/2})$.

**Step 2: term** (27)**.** Let $V_A$ solve the componentwise matrix Poisson equation

$$V_A - PV_A = A - \bar{A},$$

and set

$$\mu_{j+1} := V_A(X_{j+1}) - E[V_A(X_{j+1}) \mid X_j].$$

The finite state space makes $V_A$ and $\mu_j$ bounded. Ignoring the overall sign in (27), Poisson's equation and summation by parts give

$$\frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} \bar{A}^{-1}(V_A(X_j)\Delta_j - V_A(X_{j+1})\Delta_{j+1}) \tag{31}$$

$$+ \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} \bar{A}^{-1} V_A(X_{j+1})(\Delta_{j+1} - \Delta_j) \tag{32}$$

$$+ \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} \bar{A}^{-1} \mu_{j+1} \Delta_j. \tag{33}$$

The first sum telescopes and has mean norm $\mathcal{O}(n^{-1/2})$. Since the coefficients $A(x)$ and $b(x)$ are bounded and $\sup_j E\|\Delta_j\|^2 < \infty$,

$$E\|\Delta_{j+1} - \Delta_j\| \le C\epsilon_j,$$

so (32) has mean norm $\mathcal{O}(n^{-(\delta-1/2)})$. Finally, martingale orthogonality and boundedness of $\mu_j$ give

$$E\left\| \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} \bar{A}^{-1} \mu_{j+1} \Delta_j \right\|^2 \le \frac{C}{n} \sum_{j=0}^{n-1} E\|\Delta_j\|^2 = \mathcal{O}\left( \frac{\log n}{n^{2\delta-1}} \right),$$

where Appendix A supplies the last estimate. Hence the term in (27) satisfies

$$E\|T_{2,n}\| = \mathcal{O}\left( \frac{\sqrt{\log n}}{n^{\delta-1/2}} \right). \tag{34}$$

**Step 3: term** (28)**.** Using the same matrix Poisson equation,

$$\frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} \left( \Upsilon_j^n V_A(X_j)\Delta_j - \Upsilon_{j+1}^n V_A(X_{j+1})\Delta_{j+1} \right) \tag{35}$$

$$+ \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} \Upsilon_{j+1}^n V_A(X_{j+1})(\Delta_{j+1} - \Delta_j) \tag{36}$$

$$+ \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} (\Upsilon_{j+1}^n - \Upsilon_j^n) V_A(X_{j+1})\Delta_j \tag{37}$$

$$+ \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} \Upsilon_j^n \mu_{j+1} \Delta_j. \tag{38}$$

The first term telescopes and is $\mathcal{O}(n^{-1/2})$ in mean norm. The second is $\mathcal{O}(n^{-(\delta-1/2)})$. To handle the third, the definitions give

$$\begin{aligned}\Upsilon^n_{j+1} - \Upsilon^n_j = &\left(1 - \frac{\epsilon_j}{\epsilon_{j+1}}\right)(\Upsilon^n_{j+1} + \bar{A}^{-1}) \\ &+ \epsilon_j \bar{A}(\Upsilon^n_{j+1} + \bar{A}^{-1}) - \epsilon_j I.\end{aligned}$$

Since $|1 - \epsilon_j/\epsilon_{j+1}| = \mathcal{O}((j+1)^{-1}) = \mathcal{O}(\epsilon_j)$ and $\Upsilon^n_j$ is uniformly bounded,

$$\|\Upsilon^n_{j+1} - \Upsilon^n_j\|_{op} \le C\epsilon_j. \tag{39}$$

Thus (37) is $\mathcal{O}(n^{-(\delta-1/2)})$ in mean norm. Martingale orthogonality treats (38) exactly as (33), giving the $\sqrt{\log n}/n^{\delta-1/2}$ rate. Therefore,

$$E\|T_{3,n}\| = \mathcal{O}\left(\frac{\sqrt{\log n}}{n^{\delta-1/2}}\right). \tag{40}$$

**Step 4: term** (29)**.** The reward Poisson equation gives

$$\begin{aligned}&\frac{1}{\sqrt{n}}\sum_{j=0}^{n-1}\Upsilon^n_j r_{\mathrm{TD}}(X_j) \\ &= \frac{1}{\sqrt{n}}\sum_{j=0}^{n-1}\big(\Upsilon^n_j V_{\mathrm{TD}}(X_j) - \Upsilon^n_{j+1}V_{\mathrm{TD}}(X_{j+1})\big) \\ &\quad + \frac{1}{\sqrt{n}}\sum_{j=0}^{n-1}(\Upsilon^n_{j+1} - \Upsilon^n_j)V_{\mathrm{TD}}(X_{j+1}) \\ &\quad + \frac{1}{\sqrt{n}}\sum_{j=0}^{n-1}\Upsilon^n_j m_{j+1}.\end{aligned}$$

The first term is $\mathcal{O}(n^{-1/2})$ and, by (39), the second is $\mathcal{O}(n^{-(\delta-1/2)})$. For the martingale term, (25) gives

$$\begin{aligned}E\left\|\frac{1}{\sqrt{n}}\sum_{j=0}^{n-1}\Upsilon^n_j m_{j+1}\right\|^2 &\le \frac{C}{n}\sum_{j=0}^{n-1}\|\Upsilon^n_j\|^2_{op} \\ &= \mathcal{O}\Big(n^{-(1-\delta)}\Big).\end{aligned}$$

Hence

$$E\|T_{4,n}\| = \mathcal{O}\Big(n^{-(\delta-1/2)} + n^{-(1-\delta)/2}\Big). \tag{41}$$

**Step 5 and conclusion.** By Corollary 1 and the fact that a deterministic linear map is Lipschitz,

$$\begin{aligned}&d_{\mathcal{W}}\left(-\frac{1}{\sqrt{n}}\sum_{j=0}^{n-1}\bar{A}^{-1}r_{\mathrm{TD}}(X_j), (\bar{A}^{-1}\Sigma_\infty\bar{A}^{-T})^{1/2}Z\right) \\ &= \mathcal{O}\left(\frac{(\log n)^2}{\sqrt{n}}\right).\end{aligned}$$

For $h \in Lip_1$, successively adding the first four terms in (26)–(29) changes its expectation by at most the sum of their mean norms. Combining (34), (40), and (41) therefore gives

$$d_{\mathcal{W}}\Big(\sqrt{n}(\bar{\theta}_n - \theta^*), (\bar{A}^{-1}\Sigma_\infty \bar{A}^{-T})^{1/2} Z\Big) = \mathcal{O}\bigg(\frac{1}{\sqrt{n}} + \frac{\sqrt{\log n}}{n^{\delta-1/2}} + \frac{1}{n^{(1-\delta)/2}} + \frac{(\log n)^2}{\sqrt{n}}\bigg).$$

For every fixed $\delta \in (1/2, 1)$, $(\log n)^2/\sqrt{n} = o(\sqrt{\log n}/n^{\delta-1/2})$. This proves the stated rate. □

# 5 Conclusions

We derived a non-asymptotic central limit theorem for martingales and Markov chains, and applied the results to get rate of convergence in the central limit theorem for TD learning with averaging. The results for martingales and Markov chains may be more broadly applicable to study other reinforcement learning algorithms as well as other algorithms such as nonlinear stochastic approximation and stochastic gradient descent.

There are several possible avenues for further work:

- One can try to use the martingale CLT results in this paper to strengthen the results and allow for Markovian noise in (Anastasiou et al. 2019). However, these extensions would require us to go beyond the linear setting in this paper and consider nonlinear problems. Along similar lines, it would be also interesting to see if our results can be used to study contractive stochastic approximation as in (Chen et al. 2023a) or other stochastic approximation schemes (Borkar 2009, Kushner and Yin 2003).

- Under some conditions, the rate of convergence in the central limit theorem for i.i.d. random vectors is $\mathcal{O}(1/\sqrt{n})$ (Fang et al. 2019). It would be interesting to determine whether the $\mathcal{O}((\log n)^2/\sqrt{n})$ rate in Theorem 2 can be improved to $\mathcal{O}(1/\sqrt{n})$ under additional conditions. The Edgeworth expansion for scalar functions of Markov chains in (Kontoyiannis and Meyn 2003) suggests that the logarithmic loss might be removable. In the present proof, one factor of $(1-\beta)^{-1}$ comes from the regularity constant in Stein's equation and a second comes from summing a near-harmonic sequence after the Lindeberg decomposition. A different argument may be required to eliminate these factors.

- The approach in (Gallouët et al. 2018) that we have used is known to be not tight in terms of the dependence on the dimension $d$ for i.i.d. random vectors. Under some conditions, the dependence on $d$ can be considerably strengthened in the i.i.d. case (Courtade et al. 2019, Zhai 2018). Whether that is possible for martingales and Markov chains is an interesting question for further research.

- The Wasserstein distance used in this paper is the 1-Wasserstein distance. It would be interesting to obtain convergence rates in the stronger $p$-Wasserstein distance. Work along these lines under different conditions than in this paper can be found in (Bonis 2020), and in the previously mentioned references (Courtade et al. 2019, Zhai 2018). It would be interesting to see if such results can be obtained for martingales and Markov chains.

- It is interesting to note that, while Theorem 2 is critical to obtain convergence to the central limit theorem for TD learning, the rate of convergence is dominated by the parameter $\delta$ used in the algorithm. It would be interesting to see if the convergence rate result is tight or whether it can be improved through other techniques.

- Ignoring the polylogarithmic factor in Theorem 4, our result suggests that 2/3 is a good choice for $\delta$. This is different than the conclusion in (Haque et al. 2023), where their upper bound implies $\delta = 0.75$ is the best choice. However, these bounds ignore the constants hidden in the $\mathcal{O}$ notation, which could depend on $\delta$ (see Appendix B). Moreover, what we have is only an upper bound, so a more refined analysis is needed to understand a good choice for $\delta$.

**Acknowledgment:**

- I thank Prof. Siva Theja Maguluri, Georgia Tech, for many stimulating discussions and drawing my attention to several references during the course of this work, Prof. Sean Meyn, University of Florida, for pointing out the bias issue with fixed stepsize TD learning and why it cannot be eliminated by averaging, and other discussions, Prof. Mehrdad Moharrami, University of Iowa, for carefully reading an earlier version of the paper and providing many useful comments, and Weichen Wu, CMU, for pointing out a missing factor in Theorem 4 in an earlier version of the paper.

- I thank Professors Alexey Naumov and Sergey Samsonov for pointing out a gap in the proof of the nonasymptotic martingale CLT in an earlier version of the paper.

- Research supported by AFOSR Grant FA9550-24-1-0002, ONR Grant N00014-19-1-2566, and NSF Grants CNS 23-12714, CNS 21-06801, CCF 19-34986, and CCF 22-07547.

# References

Anastasiou A, Balasubramanian K, Erdogdu MA (2019) Normal approximation for stochastic gradient descent via non-asymptotic rates of martingale CLT. *Conference on Learning Theory*, 115–137 (PMLR).

Arnold L (1974) *Stochastic differential equations* (Wiley-InterScience).

Barbour AD (1990) Stein's method for diffusion approximations. *Probability theory and related fields* 84(3):297–322.

Benveniste A, Métivier M, Priouret P (2012) *Adaptive algorithms and stochastic approximations*, volume 22 (Springer Science & Business Media).

Bertsekas D, Tsitsiklis JN (1996) *Neuro-dynamic programming* (Athena Scientific).

Bhandari J, Russo D, Singal R (2018) A finite time analysis of temporal difference learning with linear function approximation. *Conference on learning theory*, 1691–1692 (PMLR).

Bolthausen E (1982) Exact convergence rates in some martingale central limit theorems. *The Annals of Probability* 672–688.

Bonis T (2020) Stein's method for normal approximation in wasserstein distances with application to the multivariate central limit theorem. *Probability Theory and Related Fields* 178(3-4):827–860.

Borkar V, Chen S, Devraj A, Kontoyiannis I, Meyn S (2021) The ODE method for asymptotic statistics in stochastic approximation and reinforcement learning. *arXiv preprint arXiv:2110.14427* .

Borkar VS (2009) *Stochastic approximation: a dynamical systems viewpoint*, volume 48 (Springer).

Chatterjee S (2014) A short survey of Stein's method. *arXiv preprint arXiv:1404.1392* .

Chen LH, Goldstein L, Shao QM (2010) *Normal approximation by Stein's method* (Springer Science & Business Media).

Chen Z, Maguluri ST, Shakkottai S, Shanmugam K (2023a) A Lyapunov theory for finite-sample guarantees of Markovian stochastic approximation. *Operations Research* .

Chen Z, Maguluri ST, Zubeldia M (2023b) Concentration of contractive stochastic approximation: Additive and multiplicative noise. *arXiv preprint arXiv:2303.15740* .

Courtade TA, Fathi M, Pananjady A (2019) Existence of stein kernels under a spectral gap, and discrepancy bounds. *Annales de l'Institut Henri Poincaré Probabilites et Statistiques* .

Dieuleveut A, Durmus A, Bach F (2020) Bridging the gap between constant step size stochastic gradient descent and Markov chains. *Annals of Statistics* .

Doan TT (2021) Finite-time analysis and restarting scheme for linear two-time-scale stochastic approximation. *SIAM Journal on Control and Optimization* 59(4):2798–2819.

Doan TT (2022) Nonlinear two-time-scale stochastic approximation convergence and finite-time performance. *IEEE Transactions on Automatic Control* .

Douc R, Moulines E, Priouret P, Soulier P (2018) *Markov Chains* (Springer).

Durmus A, Moulines E, Naumov A, Samsonov S (2022) Finite-time high-probability bounds for polyak-ruppert averaged iterates of linear stochastic approximation. *arXiv preprint arXiv:2207.04475* .

Fang X, Shao QM, Xu L (2019) Multivariate approximations in Wasserstein distance by Stein's method and Bismut's formula. *Probability Theory and Related Fields* 174:945–979.

Gallouët T, Mijoule G, Swan Y (2018) Regularity of solutions of the Stein equation and rates in the multivariate central limit theorem. *arXiv preprint arXiv:1805.01720* .

Glynn PW, Infanger A (2023) Solution representations for Poisson's equation, martingale structure, and the Markov chain central limit theorem. *Stochastic Systems* .

Glynn PW, Meyn SP (1996) A Liapounov bound for solutions of the Poisson equation. *The Annals of Probability* 916–931.

Gotze F (1991) On the rate of convergence in the multivariate CLT. *The Annals of Probability* 724–739.

Gupta H, Srikant R, Ying L (2019) Finite-time performance bounds and adaptive learning rate selection for two time-scale reinforcement learning. *Advances in Neural Information Processing Systems* 32.

Haque SU, Khodadadian S, Maguluri ST (2023) Tight finite time bounds of two-time-scale linear stochastic approximation with Markovian noise. *arXiv preprint arXiv:2401.00364* .

Hu B, Syed U (2019) Characterizing the exact behaviors of temporal difference learning algorithms using markov jump linear system theory. *Advances in neural information processing systems* 32.

Hu J, Doshi V, Eun DY (2024) Central limit theorem for two-timescale stochastic approximation with Markovian noise: Theory and applications. *arXiv preprint arXiv:2401.09339* .

Huo D, Chen Y, Xie Q (2023) Bias and extrapolation in Markovian linear stochastic approximation with constant stepsizes. *Abstract Proceedings of the 2023 ACM SIGMETRICS International Conference on Measurement and Modeling of Computer Systems*, 81–82.

Jain P, Kakade SM, Kidambi R, Netrapalli P, Sidford A (2018) Parallelizing stochastic gradient descent for least squares regression: mini-batching, averaging, and model misspecification. *Journal of machine learning research* 18(223):1–42.

Kontoyiannis I, Meyn SP (2003) Spectral theory and limit theorems for geometrically ergodic markov processes. *The Annals of Applied Probability* 13(1):304–362.

Kushner HJ, Yin G (2003) *Stochastic approximation and recursive algorithms and applications.(2003)* (Springer).

Lauand CK, Meyn S (2023) The curse of memory in stochastic approximation. *Proceedings of the IEEE Conference on Decision Control* (IEEE).

Li X, Liang J, Zhang Z (2023a) Online statistical inference for nonlinear stochastic approximation with Markovian data. *arXiv preprint arXiv:2302.07690* .

Li X, Yang W, Liang J, Zhang Z, Jordan MI (2023b) A statistical analysis of Polyak-Ruppert averaged Q-learning. *International Conference on Artificial Intelligence and Statistics*, 2207–2261 (PMLR).

Makowski AM, Shwartz A (2002) The Poisson equation for countable Markov chains: probabilistic methods and interpretations. *Handbook of Markov Decision Processes: Methods and Applications*, 269–303 (Springer).

Metivier M, Priouret P (1984) Applications of a Kushner and Clark lemma to general classes of stochastic algorithms. *IEEE Transactions on Information Theory* 30(2):140–151.

Meyn SP, Tweedie RL (2012) *Markov chains and stochastic stability* (Springer Science & Business Media).

Mou W, Pananjady A, Wainwright MJ, Bartlett PL (2021) Optimal and instance-dependent guarantees for Markovian linear stochastic approximation. *arXiv preprint arXiv:2112.12770* .

Moulines E, Bach F (2011) Non-asymptotic analysis of stochastic approximation algorithms for machine learning. *Advances in neural information processing systems* 24.

Nagaraj D, Wu X, Bresler G, Jain P, Netrapalli P (2020) Least squares regression with Markovian data: Fundamental limits and algorithms. *Advances in neural information processing systems* 33:16666–16676.

Polyak BT (1990) A new method of stochastic approximation type. *Avtomatika i telemekhanika* 98–107.

Polyak BT, Juditsky AB (1992) Acceleration of stochastic approximation by averaging. *SIAM journal on control and optimization* 30(4):838–855.

Röllin A (2018) On quantitative bounds in the mean martingale central limit theorem. *Statistics & Probability Letters* 138:171–176.

Ross N (2011) Fundamentals of Stein's method. *Probability Surveys* .

Roy A, Balasubramanian K (2023) Online covariance estimation for stochastic gradient descent under Markovian sampling. *arXiv preprint arXiv:2308.01481* .

Ruppert D (1988) Efficient estimations from a slowly convergent Robbins-Monro process. Technical report, Cornell University Operations Research and Industrial Engineering.

Srikant R, Ying L (2019) Finite-time error bounds for linear stochastic approximation and TD learning. *Conference on Learning Theory*, 2803–2830 (PMLR).

Stein C (1972) A bound for the error in the normal approximation to the distribution of a sum of dependent random variables. *Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability, Volume 2: Probability Theory*, volume 6, 583–603 (University of California Press).

Sutton RS, Barto AG (2018) *Reinforcement learning: An introduction* (MIT press).

Telgarsky M (2022) Stochastic linear optimization never overfits with quadratically-bounded losses on general data. *Conference on Learning Theory*, 5453–5488 (PMLR).

Tsitsiklis J, Van Roy B (1996) Analysis of temporal-difference learning with function approximation. *Advances in neural information processing systems* 9.

Zhai A (2018) A high-dimensional CLT in $W_2$ distance with near optimal convergence rate. *Probability Theory and Related Fields* 170:821–845.

# A Properties of $E(||\Delta_k||^2)$

In this section and the next, constants are local to the appendix; letters used for constants elsewhere in the paper may therefore be reused. Since the Markov chain is finite-state, irreducible, and aperiodic, its mixing time satisfies $\tau_\epsilon = O(\log(1/\epsilon))$; hence, for $\epsilon_k = (k+1)^{-\delta}$, we have $k - \tau_{\epsilon_k} \geq 0$, $\epsilon_{k-\tau_{\epsilon_k}}/\epsilon_k = O(1)$, and $\epsilon_k \tau_{\epsilon_k} \to 0$, so the diminishing-stepsize conditions of (Srikant and Ying 2019, Theorem 12) hold for all sufficiently large $k$. Thus, there exists a positive integer $k^*$ such that for any $k \geq k^*$, we have

$$E\left[\|\Delta_k\|^2\right] \leq \kappa_1 \left(\prod_{j=k^*}^{k-1} a_j\right) + \kappa_2 \sum_{j=k^*}^{k-1} b_j \left(\prod_{l=j+1}^{k-1} a_l\right),$$

where $a_j = 1 - \kappa_3 \epsilon_j$, $b_j = \kappa_4 \epsilon_j^2 \log(1/\epsilon_j)$, and $\kappa_1, \kappa_2, \kappa_3, \kappa_4$ are positive constants. Since $a_j \leq \exp(-\kappa_3 \epsilon_j)$, the first term is bounded by $C \exp(-ck^{1-\delta})$ and hence goes to zero at a stretched-

exponential rate. The second term can be upper bounded as follows:

$$\begin{aligned}
&\kappa_2\kappa_4 \sum_{j=0}^{k-1} \epsilon_j^2 \log\left(\frac{1}{\epsilon_j}\right) \exp\left(-\kappa_3 \sum_{l=j+1}^{k-1} \epsilon_l\right) \\
&\leq \kappa_2\kappa_4\delta \log k \sum_{j=0}^{k-1} \epsilon_j^2 \exp\left(-\kappa_3 \sum_{l=j+1}^{k-1} \epsilon_l\right) \\
&\leq \kappa_2\kappa_4\delta \log k \left(\sum_{j=0}^{\lceil k/2\rceil} \exp\left(-\kappa_3 \sum_{l=j+1}^{k-1} \epsilon_l\right) + \sum_{j=\lceil k/2\rceil+1}^{k-1} \epsilon_j^2\right) \\
&= \kappa_2\kappa_4\delta \log k \left(\sum_{j=0}^{\lceil k/2\rceil} \exp\left(-\kappa_3 \sum_{l=j+1}^{k-1} (\frac{1}{l+1})^{\delta}\right) + \sum_{j=\lceil k/2\rceil+1}^{k-1} (\frac{1}{j+1})^{2\delta}\right) = \mathcal{O}\left(\frac{\log k}{k^{2\delta-1}}\right).
\end{aligned}$$

# B Properties of $\Upsilon_j^n$ and $\Phi_j^n$

Recall the definition of $\Upsilon_j^n$ and $\Phi_j^n$ in (23). In this section of the appendix, we examine certain expressions which are shown to be either bounded or converging to zero in (Polyak and Juditsky 1992), and derive explicit bounds for them as functions of $\delta$.

From (Polyak and Juditsky 1992, page 846), we know that $\Phi_j^t$ can be written as

$$\Phi_j^t = C_j^t + \bar{A}^{-1} D_j^t,$$

where

$$\|C_j^t\|_{op} \leq \frac{\epsilon_j - \epsilon_{j+1}}{\epsilon_j} \sum_{i=j}^{t} m_j^i e^{-\lambda m_j^i}, \qquad D_j^t = \prod_{l=j}^{t-1} (I - \epsilon_l \bar{A}),$$

and

$$\|D_j^t\|_{op} \leq K e^{-\alpha m_j^{t-1}}, \qquad m_j^i := \sum_{k=j}^{i} \epsilon_k,$$

for $t \geq j+1$ and some constants $K, \alpha, \lambda > 0$.

We first bound the contribution of $D_j^t$. Since $\epsilon_k = (k+1)^{-\delta}$ is nonincreasing, for $0 \leq j \leq t-1$,

$$m_j^{t-1} = \sum_{k=j}^{t-1} \epsilon_k \geq (t-j)t^{-\delta}.$$

Hence

$$\|D_j^t\|_{op}^2 \leq K^2 \exp\Big(-2\alpha(t-j)t^{-\delta}\Big).$$

Therefore,

$$\begin{aligned}
\sum_{j=0}^{t-1} \|D_j^t\|_{op}^2 &\leq K^2 \sum_{r=1}^{t} \exp\Big(-2\alpha r t^{-\delta}\Big) \\
&\leq \frac{K^2}{1 - \exp(-2\alpha t^{-\delta})} = \mathcal{O}(t^{\delta}),
\end{aligned}$$

and consequently

$$\frac{1}{t}\sum_{j=0}^{t-1}\|D_j^t\|_{op}^2 = \mathcal{O}\left(\frac{1}{t^{1-\delta}}\right). \tag{42}$$

This rate is natural: for indices $j$ within distance of order $t^\delta$ from $t$, the product $D_j^t$ contains only a small number of factors and need not yet be small.

We next bound $C_j^t$. Using

$$\frac{\epsilon_j - \epsilon_{j+1}}{\epsilon_j} = 1 - \left(\frac{j+1}{j+2}\right)^\delta \le \frac{1}{j+2},$$

we have

$$\|C_j^t\|_{op} \le \frac{1}{j+2}\sum_{i=j}^{t} m_j^i e^{-\lambda m_j^i}.$$

Fix $\mu > \delta/\lambda$. Since

$$\epsilon_k = (k+1)^{-\delta} \ge \frac{\mu}{k+1}$$

for all sufficiently large $k$, there exists an integer $K_{\mu,\delta}$ such that, for all $i \ge j \ge K_{\mu,\delta}$,

$$m_j^i = \sum_{k=j}^{i}\epsilon_k \ge \mu\sum_{k=j}^{i}\frac{1}{k+1} \ge \mu\log\frac{i+1}{j+1}.$$

Thus,

$$\left(\frac{i+1}{j+1}\right)^\delta \le \exp\left(\frac{\delta}{\mu}m_j^i\right).$$

Using $m_j^i - m_j^{i-1} = \epsilon_i = (i+1)^{-\delta}$, we obtain, for $j \ge K_{\mu,\delta}$,

$$\begin{aligned}
\|C_j^t\|_{op} &\le \frac{1}{j+2}\sum_{i=j}^{t} m_j^i e^{-\lambda m_j^i}\frac{m_j^i - m_j^{i-1}}{\epsilon_i} \\
&= \frac{(j+1)^\delta}{j+2}\sum_{i=j}^{t} m_j^i e^{-\lambda m_j^i}\left(\frac{i+1}{j+1}\right)^\delta (m_j^i - m_j^{i-1}) \\
&\le \frac{(j+1)^\delta}{j+2}\sum_{i=j}^{t} m_j^i \exp\left(-\left(\lambda - \frac{\delta}{\mu}\right)m_j^i\right)(m_j^i - m_j^{i-1}) \\
&\le \frac{C}{(j+1)^{1-\delta}},
\end{aligned}$$

where $C < \infty$ depends on $\delta, \lambda, \mu$, but not on $j$ or $t$. Enlarging the constant to absorb the finitely many indices $j < K_{\mu,\delta}$, we obtain

$$\|C_j^t\|_{op} \le \frac{C}{(j+1)^{1-\delta}} \qquad \text{for all } 0 \le j \le t-1. \tag{43}$$

It follows that

$$\begin{aligned}
\frac{1}{t}\sum_{j=0}^{t-1}\|C_j^t\|_{op}^2 &\le \frac{C^2}{t}\sum_{j=0}^{t-1}\frac{1}{(j+1)^{2(1-\delta)}} \\
&= \mathcal{O}\left(\frac{1}{t^{2(1-\delta)}}\right) = \mathcal{O}\left(\frac{1}{t^{1-\delta}}\right),
\end{aligned} \tag{44}$$

where the final, weaker bound is sufficient for our purposes.

Combining

$$\Phi_j^t = C_j^t + \bar{A}^{-1} D_j^t$$

with (42) and (44), we obtain

$$\frac{1}{t}\sum_{j=0}^{t-1} \|\Phi_j^t\|_{op}^2 = \mathcal{O}\left(\frac{1}{t^{1-\delta}}\right). \tag{45}$$

Moreover, (43) and the uniform bound on $D_j^t$ imply

$$\sup_{t\geq 1} \max_{0\leq j\leq t-1} \|\Phi_j^t\|_{op} < \infty. \tag{46}$$

Finally, we transfer this estimate from $\Phi_j^n$ to $\Upsilon_j^n$. From the definitions in (23),

$$\begin{aligned}
\Upsilon_j^n &= \epsilon_j \sum_{k=j+1}^{n} \prod_{l=j+1}^{k-1} (I - \epsilon_l \bar{A}) - \bar{A}^{-1} \\
&= \frac{\epsilon_j}{\epsilon_{j+1}} \left(\Phi_{j+1}^{n+1} + \bar{A}^{-1}\right) - \bar{A}^{-1} \\
&= \frac{\epsilon_j}{\epsilon_{j+1}} \Phi_{j+1}^{n+1} + \left(\frac{\epsilon_j}{\epsilon_{j+1}} - 1\right) \bar{A}^{-1}.
\end{aligned}$$

Since

$$\frac{\epsilon_j}{\epsilon_{j+1}} = \left(\frac{j+2}{j+1}\right)^{\delta}$$

is uniformly bounded and

$$\frac{\epsilon_j}{\epsilon_{j+1}} - 1 = \mathcal{O}\left(\frac{1}{j+1}\right),$$

(46), together with $\Upsilon_n^n = -\bar{A}^{-1}$, gives

$$\sup_{n\geq 1} \max_{0\leq j\leq n} \|\Upsilon_j^n\|_{op} < \infty. \tag{47}$$

It also follows from (45) that

$$\begin{aligned}
\frac{1}{n}\sum_{j=0}^{n-1} \|\Upsilon_j^n\|_{op}^2 &\leq \frac{C}{n}\sum_{j=0}^{n-1} \|\Phi_{j+1}^{n+1}\|_{op}^2 + \frac{C}{n}\sum_{j=0}^{n-1} \frac{1}{(j+1)^2} \\
&= \mathcal{O}\left(\frac{1}{n^{1-\delta}}\right).
\end{aligned}$$

This proves both estimates stated in (25).